

## C++ Intermediate Test

Test Reference No: PLMAcademy-Nov2025-General-Batch

Total Marks : 30

Test Duration : 1 hour

### Instructions to Candidate

1. All questions are compulsory/mandatory.
2. Do not write anything on the question paper.  
Write everything on the answer sheet provided.
3. This is a closed book test.
4. There is no negative marking.
5. Wherever there is ambiguity, make suitable assumptions and proceed.  
Document the assumptions in the answer.
6. Assume C++ 11 language standard.

Assume usage of Microsoft Visual Studio development platform.

| Question No. | Question                                                                                                                                                                                                                                              | Marks |
|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 1            | What is an abstract class. Explain with minimal code.                                                                                                                                                                                                 | 1     |
| 2            | Will the following code compile and execute? If so, what will be the output?<br><pre>#include &lt;iostream&gt; using namespace std;  void f() const {     cout &lt;&lt; "In function f" &lt;&lt; endl; }  int main() {     f();     return 0; }</pre> | 1     |
| 3            | Write the declaration for the following:<br>a. Array of 5 integers<br>b. Pointer to an integer                                                                                                                                                        | 2     |

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|   | c. Pointer to an array of 5 integers<br>d. Array of 5 pointer to integers                                                                                                                                                                                           |   |
| 4 | What does the new and delete operator do? Explain with minimal code and memory diagram.                                                                                                                                                                             | 2 |
| 5 | Will the following code compile and link?<br><br><pre> #define Begin { #define End   }  int main ( ) Begin     return 0 ; End </pre>                                                                                                                                | 1 |
| 6 | Consider the following code:<br><br><pre> class X { private:     int m_x, m_y ;  public:     X( int x, int y ) : m_x( x ), m_y ( y ) {} }; </pre> What does the following line do in the above code:<br><br><pre> X( int x, int y ) : m_x( x ), m_y ( y ) {} </pre> | 1 |
| 7 | Write the code in the fill in the blank so that the code will compile and link, and it will work properly.<br><br><pre> class X { private:     int m_i ;     _____ };  int main ( ) { </pre>                                                                        | 1 |

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|   | <pre> X x1 ; x1.m_i = 10 ; return 0 ; } </pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| 8 | <p>Write the code in the fill in the blank so that this code will compile and run, and the function f will be called.</p> <pre> #include &lt;iostream&gt; using namespace std ;  void f ( ) {     cout &lt;&lt; "Function f called" &lt;&lt; endl ; }  void g( _____ ) {     (*fp) ( ) ; }  int main ( ) {     g( &amp;f ) ;     return 0 ; } </pre>                                                                                                                                                   | 1 |
| 9 | <p>What will be the output of the below program:</p> <pre> #include&lt;iostream&gt; using namespace std;  class CWindow { public:     CWindow() { cout &lt;&lt; "CWindow constructor called" &lt;&lt; endl; }     ~CWindow() { cout &lt;&lt; "CWindow destructor called" &lt;&lt; endl; } };  class CDialog :public CWindow { public:     CDialog() { cout &lt;&lt; "CDialog constructor called" &lt;&lt; endl; }     ~CDialog() { cout &lt;&lt; "CDialog destructor called" &lt;&lt; endl; } } </pre> | 1 |

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|    | <pre> };  class CPopupWindow :public CDialog { public:     CPopupWindow() { cout &lt;&lt; "CPopupWindow constructor called" &lt;&lt; endl; }     ~CPopupWindow() { cout &lt;&lt; "CPopupWindow destructor called" &lt;&lt; endl; } };  int main() {     CPopupWindow* pw = new CPopupWindow();     delete pw; } </pre>         |   |
| 10 | <p>What will be the output of the below program:</p> <pre> #include &lt;iostream&gt; using namespace std;  int main() {     float arr[5] = {12.5, 10.0, 13.5, 90.5, 0.5};      float *ptr1 = &amp;arr[0];     float *ptr2 = ptr1 + 3;      cout&lt;&lt;*ptr2&lt;&lt;" ";     cout&lt;&lt; ptr2 - ptr1;      return 0; } </pre> | 1 |
| 11 | <p>What will be the output of below program?</p> <pre> #include &lt;iostream&gt; using namespace std;  int main() {     char *ptr;     char Str[] = "abcdefg"; </pre>                                                                                                                                                          | 2 |

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|    | <pre> ptr = Str; ptr += 5;  cout &lt;&lt; ptr;  return 0; } </pre>                                                                                                                                                                                                                                                                                                                  |   |
| 12 | <p>What is the command to compile the file a.cpp, into an a.obj.</p> <p>What is the command to get the symbol table from an object file?</p> <p>What will be the important/significant symbol table entries for the following code with respect to the functions in the following code:</p> <pre> void f ( );  void g() { }  int main ( ) {     f( );     g( ); } File a.cpp </pre> | 2 |
| 13 | Is dangling pointer same as dangling reference? If not explain with minimal code example for each.                                                                                                                                                                                                                                                                                  | 2 |
| 14 | Can a constructor be private? If yes, how does one go about creating objects of this class? Write suitable code to explain the same.                                                                                                                                                                                                                                                | 3 |
| 15 | <p>Show the memory layout for the various local variables, i, pi, ppi.</p> <p>Make suitable assumptions related to the memory addresses.</p> <p>Show 32 bit addresses in hexadecimal format.</p>                                                                                                                                                                                    | 1 |

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|    | <pre> int main( ) { int i = 100 ; int *pi = &amp;i ; int **ppi = &amp;pi ;      return 0 ; } </pre>                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| 16 | <p>Will the following code compile?<br/>If so, what will be the output? If not mention the reason behind the error.</p> <pre> #include &lt;iostream&gt; using namespace std;  class Shape { public:     void f(int i) { cout &lt;&lt; "In function Shape::f " &lt;&lt; i &lt;&lt; endl; }     void f() { cout &lt;&lt; "In function Shape::f " &lt;&lt; endl; } };  class Circle :public Shape { public:     void f(){ cout &lt;&lt; "In function Circle::f " &lt;&lt; endl; } };  int main() {     Circle c;     c.f(10);     return 0; } </pre> | 1 |
| 17 | <p>Explain call by value, call by pointer/address and call by reference with a suitable example.</p> <p>Write minimal code example with line numbers.</p> <p>Explain with memory diagram at significant line numbers, i.e. explain with memory diagram after important line numbers.</p>                                                                                                                                                                                                                                                          | 2 |
| 18 | In C++, how is encapsulation done. Show with a suitable example.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 2 |

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| 19 | <p>In the following code, the pointer to integer is on stack.</p> <pre>void f( )<br/>{<br/>  int *pi;<br/>}</pre> <p>Write a piece of code, which demonstrates that the pointer to integer is somehow created on heap, i.e. the pointer to integer is on heap.</p> | 1 |
| 20 | <p>Write a program to return multiple values from a function.</p>                                                                                                                                                                                                  | 2 |